Information Retrieval With Chessboard-Shaped Topology for Hyperspectral Target Detection

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***Abstract*— Given a priori knowledge, hyperspectral target detection aims to locate objects of interest within specific scenes by utilizing differences in spectral characteristics among various land covers. However, for those traditional model-driven detectors with monotonic analytical mode, they perform mediocrely in the disassembly of hyperspectral image (HSI) data, failing to cope with real scenes with complexity. The discrepancy between fixed model assumptions and HSI data severely reduces detection effects, leading to the inability of such methods to mine deep-level features and adapt to the variability of imaging scenes. To overcome the limitations of traditional methods, we propose a chessboard-shaped topological framework for high-dimensional data structures to disassemble an HSI from both spatial and spectral dimensions adaptively. With hyperspectral target detection refined into an information retrieval task in a topological space, a chessboard-shaped topology for hyperspectral target detection (CTTD) is proposed. In the topological space, latent and hidden data features of original images are presented in an intuitive way. Therefore, the differences in both spatial and spectral dimensions between the two classes of objects, namely, target and background, are specifically amplified and exploited to perform the information retrieval task with superior performance. Extensive experimental results on benchmark HSI datasets demonstrate that CTTD can efficiently adapt to the variability of real scenes while extracting abundant and detailed information for accurate target localization. Moreover, both detection effects and computational efficiency exhibited by the proposed method provide strong support for its popularization in practical applications.**

***Index Terms*— Hyperspectral image (HSI), information retrieval, machine learning, target detection, topology.**

Manuscript received 27 March 2023; revised 18 May 2023; accepted 6 June 2023. Date of publication 9 June 2023; date of current version 21 June 2023. This work was supported by the National Natural Science Foundation of China under Grant 62161160336 and Grant 42030111. *(Corresponding author: Lianru Gao.)*

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Digital Object Identifier 10.1109/TGRS.2023.3284653

I. Introduction

TYPERSPECTRAL imaging, as one of the research hotspots and frontier directions of remote sensing technology, has made significant achievements over the past three decades [1], [2], [3]. With exhibiting unique advantages in a variety of fields covering civilian and military, hyperspectral data have been popularized and widely used [4], [5], [6]. Materials reflect, absorb, and emit electromagnetic energy at specific wavelengths, and such property is associated with their physical structure and chemical composition [7], [8]. Based on this, hyperspectral sensors (i.e., imaging spectrometers) acquire digital images in continuous and subdivided spectral bands spanning visible light, near infrared, and short-wave infrared (0.4–2.5 μm) [9], [10], which can provide an essentially continuous radiation spectrum for each pixel contained in a specific scene [11], [12]. Hyperspectral target detection is an important application exploiting hyperspectral image (HSI) data, enabling the localization of targets of interest based on spectral characteristics [13], [14]. Detection algorithms based on statistics, signal processing, machine learning, and other enlightening disciplines are rapidly emerging [15], [16]. Currently, there are well-established and mature design processes and algorithmic systems developed based on various mechanisms [17], [18], [19].

With the availability of priori knowledge of target, hyperspectral target detection is capable of locating objects of interest in a specific scene with high accuracy [1], [20]. Supervised traditional detectors can be mainly divided into three categories [18]. The first is simple matching algorithm. Common waveform matching algorithms can be further subdivided according to various matching criteria, such as distance, angle, similarity coefficient, and encoding between two spectral vectors [21]. They are represented by algorithms such as spectral minimum distance, spectral angle mapping (SAM), cross correlation spectral matching (CCSM), and binary encoding matching [1]. The second category estimates the information of land covers through statistics such as correlation matrix and covariance matrix of HSI data to implement target detection [22]. Representative algorithms include constrained energy minimization (CEM) [23], adaptive coherence/cosine estimator (ACE) [24], adaptive matched filter (AMF) [25], and generalized likelihood ratio test (GLRT) [26], [27]. The third category performs endmember extraction of images where targets of interest exist in the form of subpixels by means of hyperspectral unmixing techniques [9], [28], attempting to

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transform this situation into one where both the target and the background priori information are known for processing by acquiring background information [29]. In addition, for cases where background information is available or estimated beforehand, orthogonal subspace projection (OSP) and its various derivative versions [30], as well as target-constrained interference minimized filter (TCIMF) can detect objects of interest with low false alarm rates (FARs) by suppressing background features [31].

Most of the aforementioned traditional methods, relying on specific model assumptions, have relatively low computational complexity and can achieve an effective target-background separation when an imaging scene is relatively homogeneous in distribution. However, with the variability of scenes and the development of hyperspectral imaging techniques, the quality of HSI data has improved significantly. The effectiveness of traditional methods is limited due to the fact that fixed model assumptions may not be able to fit real data in various complex scenes. In this case, machine learning-based methods stand out for their ability to tap into the wealth of detailed information contained in HSIs and are more applicable to real data with complexity. Several machine learning-based techniques have gained significant attention, including sparse representation [32], kernel methods [33], discriminative subspace analysis [34], self-learning of spectral data [35], manifold learning [36], tensor decomposition [37], and deep learning [38]. In recent years, a great number of machine learning-based methods have emerged in the field of hyperspectral target detection and achieved impressive performance. For example, representation-based methods, such as sparse representation-based detector (SRD) [32], combined sparse and collaborative representation-based detector (CSCR) [39], and sparse and low-rank matrix decomposition (SLRMD) [40], achieve a more accurate separation by constructing various forms of target or background dictionaries to represent testing pixels. Kernel-OSP [41], kernel matched subspace detector (Kernel-MSD) [33], and kernel TCIMF (Kernel-TCIMF) employ various kernel mappings to equip detectors with processing capability for nonlinear data [42]. Methods, such as tensor MSD (TMSD) and canonical polyadic matched filter (CPMF), improve the separability of target and background by modeling an HSI data cube as a higher order tensor to exploit both spatial and spectral information [43], [44]. In addition, deep learning has grown significantly as an important branch of machine learning. The massive amount of algorithms derived from convolutional neural network (CNN) [45], autoencoder (AE) [46], deep belief network (DBN) [47], and generative adversarial network (GAN) continues to drive the development of high-precision detection [38], [48]. In summary, the migration of machine learning theories in hyperspectral target detection has yielded various representative algorithms with excellent performance, exhibiting strong parsing ability for real data with complexity. However, such machine learning-based methods encounter difficulties in implementing target detection in HSIs with insufficient training samples, sensitivity to parameters, and high computational complexity. Although the stringent validity conditions prevent machine learning-based methods from fully demonstrating their advantages in practical applications, various theoretical migrations have inspired more possibilities for the design of detectors.

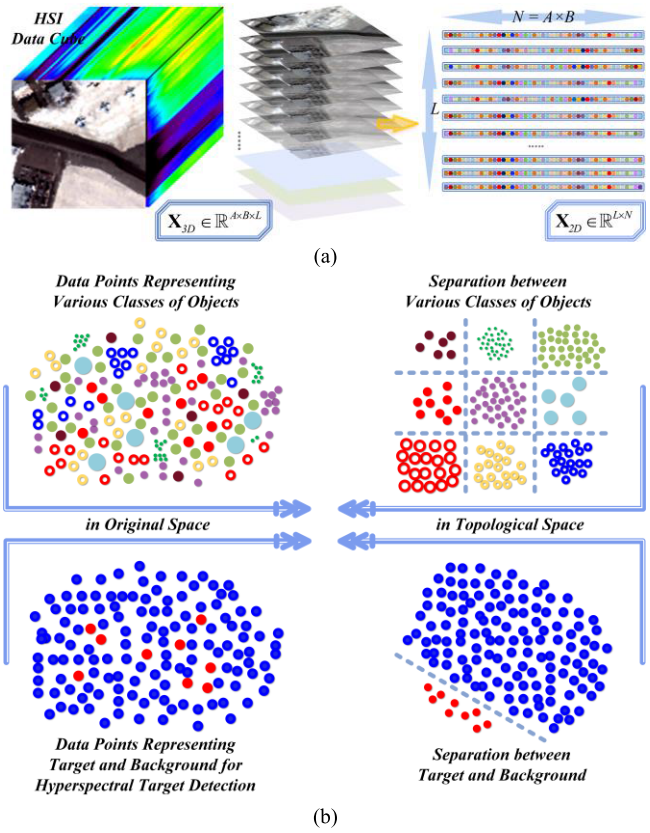


Fig. 1. Illustration of the application of point-set topology to an information retrieval task. (a) Different forms of HSI data. (b) Disassembly of data from disorder to order.

In order to break free from the constraints of traditional model assumptions and overcome the limitations caused by the stringent validity conditions, this article finds the solutions for hyperspectral target detection based on data features of HSIs themselves rather than specific model assumptions. Given a query, information retrieval is the process of finding relevant information in an information collection by users with the help of certain methods or retrieval tools [49]. As a powerful mathematical tool, theories of point-set topology can be fully applied in data mining tasks such as information retrieval and provide feasible solutions [50], [51], [52]. Specifically, a dataset corresponding to the information collection being processed is mostly chaotic and disordered, which is not conducive to the adoption of specific methods or tools to meet the retrieval requirements. The idea behind the application of point-set topology is to map the original data into a certain number of point sets, whose relationships are intuitively represented in the topological space. Based on this, the evaluation of data in an information collection can be implemented according to specific task requirements. The essential purpose of this process is to disassemble chaotic and disordered data into a neat and ordered form, which facilitates the subsequent extraction of useful information for a particular application, as shown in Fig. 1.

In this article, the proposed method starts with the construction of a framework in the form of a chessboard to disassemble an HSI being processed. The process of disassembly generates a chessboard-shaped topological space, which consists of a certain number of point sets. The number of elements contained in each point set is defined as its cardinality. The two classes of pixels, target and background, are inevitable

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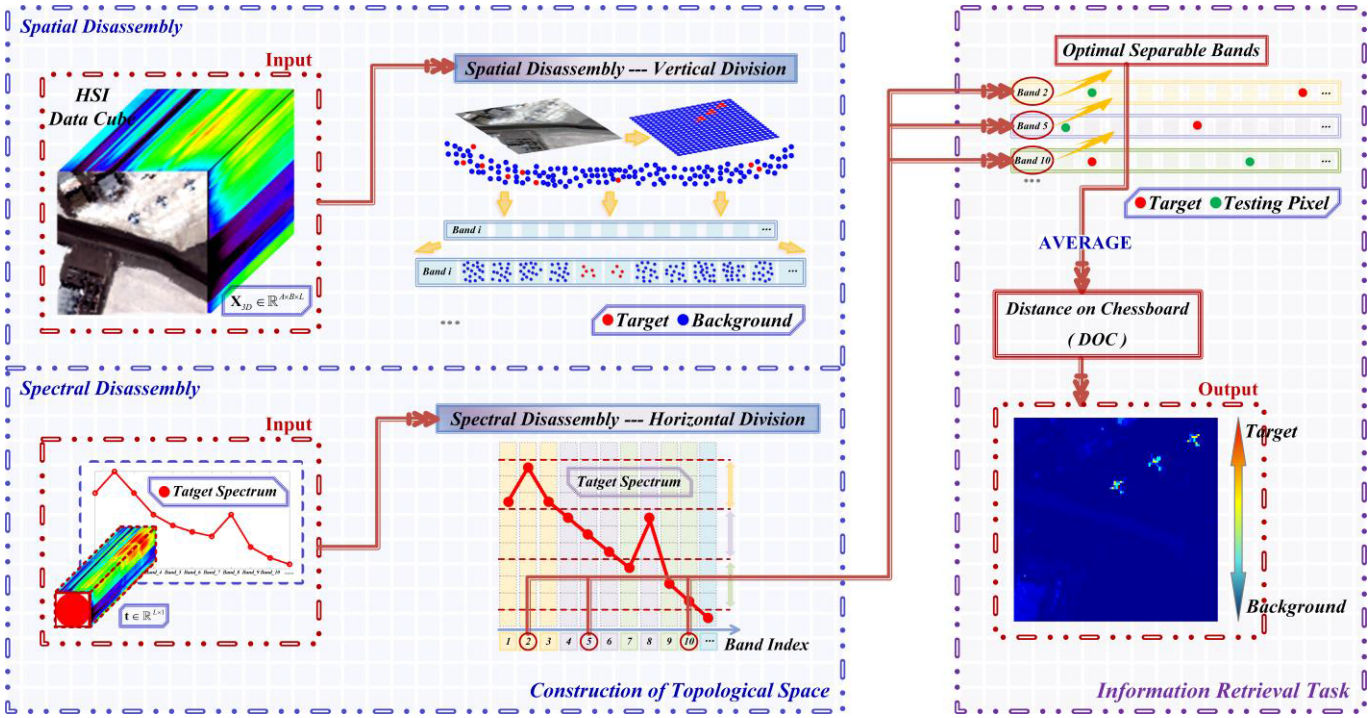


Fig. 2. Flowchart of the proposed CTTD for hyperspectral target detection.

to be divided into separate point sets due to differences in quantity, spatial distribution, and spectral characteristics. In the topological space, differences in relative position and density of these two classes are directly reflected in topological relationships between the separate point sets. As a result, latent and hidden data features in the original space are transformed into intuitive properties in the topological space, which significantly reduces the difficulty of retrieving targets of interest for information retrieval tasks. On this basis, with a given target spectrum as reference information, the algorithmic output for the core requirement of highlighting the target while suppressing the background can provide an effective detection scheme for HSIs. The contributions of related research work can be briefly summarized as follows.

1) Hyperspectral target detection is refined into an information retrieval task and the theory of point-set topology is adopted to design a detection scheme, thus completely breaking free from the constraints of the traditional monotonous analytical mode based on fixed model assumptions. Retrieval based on data features of HSIs themselves equips the proposed chessboard-shaped topology for hyperspectral target detection (CTTD) with adaptability to the variability of real scenes.
2) For high-dimensional data structures of HSIs, a chessboard-shaped framework is designed specifically to enable both spatial and spectral disassembly through vertical and horizontal divisions, whereby a topological space is generated. In such a space, differences in quantity, spatial distribution, and spectral characteristics between various classes of objects are amplified and presented in an intuitive way so as to ensure an accurate target-background separation.
3) Based on a given priori knowledge, the optimal separable bands between the target and the background are adaptively selected for an HSI using the chessboard topology to develop a measurement of separation called "distance on chessboard" for the detection output. By overcoming the inherent spectral redundancy of HSIs while improving computational efficiency, the proposed CTTD demonstrates a highly competitive comprehensive performance.

This article is organized as follows. Section II provides a detailed methodological description of the proposed CTTD. Section III gives the experimental results, analysis, and discussion. Section IV draws the conclusions on the research work in this article.

II. Information Retrieval With Chessboard-Shaped Topology for Hyperspectral Target Detection (CTTD)

The detection process of our method CTTD is shown in Fig. 2. In Sections II-A–II-C, the implementation of CTTD is described in detail in three parts: 1) constructing a chessboard-shaped framework by disassembling an HSI in both spatial and spectral dimensions to generate a topological space for target-background separation; 2) developing a separation measurement for the detection output by selecting the optimal separable bands with the chessboard topology; and 3) performing an information retrieval task in the topological space to accomplish hyperspectral target detection.

A. Construction of Chessboard-Shaped Topological Framework

With the given target spectrum treated as a query and the HSI being processed as an information collection, target detection in HSIs can be refined into an information retrieval task that aims to find pixels belonging to the target class. In the original hyperspectral dataset, pixels are arranged in a chaotic and disordered form, which is not conducive to

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designing or employing specific methods and tools to meet retrieval requirements. To overcome this challenge, a point-set topology is adopted to provide theoretical support. The original data are mapped into a certain number of point sets through a specific form of framework to generate a topological space. In the topological space, all data are contained in the point sets, which are arranged in a neat and ordered form. Also, the relevance of the data points to the given query is reflected in intuitive properties. The transformation of data from disorder to order greatly reduces the difficulty of retrieval and is quite conducive to finding data points of the same class as the given query. Given that the form of the topological framework directly determines the generation of the topological space, it is crucial that the framework is constructed to meet the detection requirements for HSIs.

For HSIs, achieving an accurate target-background separation provides a basis for subsequent high-precision detection. An HSI exists in the form of a 3-D data cube, which can be treated as a collection of images in specific bands or as a collection of spectra at specific locations; hence, the differences between target and background are reflected in both spatial and spectral dimensions. Inspired by these two forms of data observation, a chessboard-shaped topological framework is devised by performing a spatial disassembly of an image in a certain band and a spectral disassembly of a specific spectrum.

Spatial disassembly corresponds to the vertical division of a chessboard. Assume that $\mathbf{X} \in \mathbb{R}^{L\times N}$ is the HSI being processed, L denotes the number of bands, and N denotes the number of pixels. An image in a specific band of $\mathbf{X}$ is extracted to obtain $\mathbf{X}_i \in \mathbb{R}^{1\times N}, i = 1, 2, \ldots, L$, and its minimum–maximum interval is divided equally into ver parts, which correspond to ver columns. For each pixel of the HSI in the current band, it is assigned to the corresponding column according to the interval in which its value lies. Each column corresponds to a grid-shaped point set, and the number of pixels contained in it is its cardinality. All bands of the HSI are traversed and the above operation is repeated to achieve a vertical division for the chessboard, as shown in Fig. 3. In each band, the sum of the cardinalities of all ver columns is equal to the total number of pixels of the HSI, N. Simultaneously, for a given target spectrum $\mathbf{t} \in \mathbb{R}^{L\times 1}$, the column to which $\mathbf{t}$ is assigned is tracked according to the interval in which its value lies in each band. Also, the cardinalities of the corresponding grid-shaped point sets on all bands are recorded in $\mathbf{Card}_t = [\mathrm{Card}_1, \ldots, \mathrm{Card}_L]^{\mathrm{T}} \in \mathbb{R}^{L\times 1}$.

Spectral disassembly corresponds to the horizontal division of a chessboard. For a given target spectrum $\mathbf{t}$, its minimum–maximum interval is divided equally into hor parts, which correspond to hor rows. For each band of this target spectrum, it is assigned to the corresponding row according to the interval in which its value lies, as shown in Fig. 4. Each row records a set of band indexes, and the sum of the number of the band indexes in all rows is equal to the total number of bands of the HSI, L.

It is worth noting that the priori information about the target is not utilized in the described above spatial disassembly. Hence, it cannot be guaranteed that the target and the background pixels will be completely separated. As shown in Fig. 3, there are cases where both classes of pixels exist in the same grid. However, the abovementioned cases are rare.



Fig. 3. Spatial disassembly corresponding to the vertical division of the chessboard.

In most cases, the target and the background are distributed within distinguishable grid-shaped point sets, which differ significantly in quantity and density. Such a fact provides a solid and reliable basis for accurate separation of targets from the background. Meanwhile, in the spectral disassembly, the given target spectrum is fully utilized. With the spatial and spectral disassembly, the vertical and horizontal divisions are integrated to construct a chessboard-shaped topological framework. A toy example is shown in Fig. 5, where for the given $\mathbf{t}$, the corresponding vertical divisions are found according to the band indexes in each row. In the first row of the chessboard, the recorded band indexes corresponding to $\mathbf{t}$ are Band_1, Band_2, and Band_3, whose vertical divisions are marked in yellow. In the second row, the band indexes of $\mathbf{t}$ are Band_4, Band_5, Band_6, and Band_8, whose vertical divisions are marked in purple. In the third row, the band indexes of $\mathbf{t}$ are Band_7, Band_9, and Band_{10}, whose vertical divisions are marked in green. Analogously, from the first row to the hor−row, a chessboard-shaped frame is constructed to generate the corresponding topological space.

For an HSI data cube, the differences in data features between various classes of objects are essentially numerical differences. Thus, the chessboard-shaped framework is constructed based entirely on the magnitude of values on the spatial and spectral dimensions. In the chessboard-shaped topological space, all pixels of the original HSI are rearranged from disorder to order through the spatial and spectral disassembly. For instance, in Fig. 3, the red pixels represent the target, while blue pixels represent the background. In a given band, if the center values corresponding to each of two adjacent columns span the same interval, the pixels in the same

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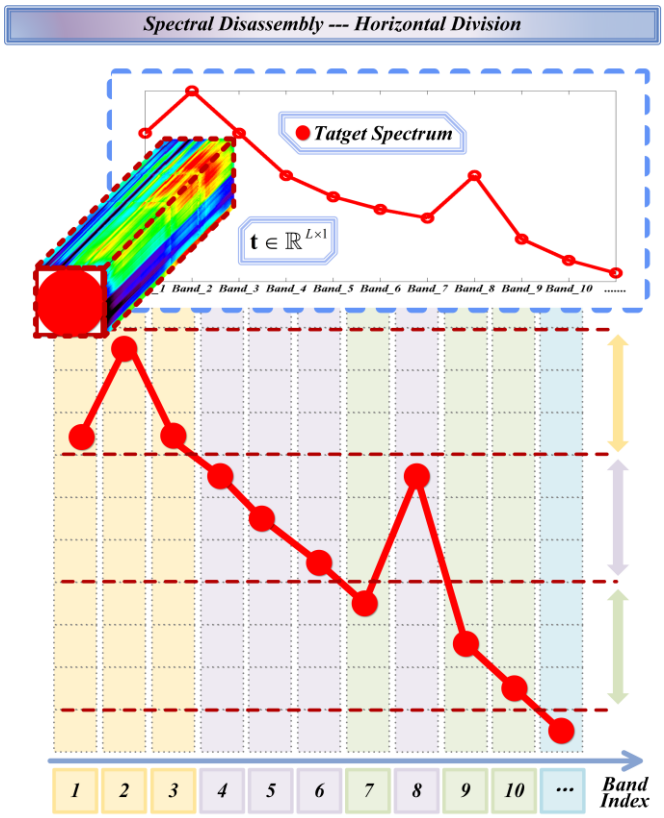


Fig. 4. Spectral disassembly corresponding to the horizontal division of the chessboard.

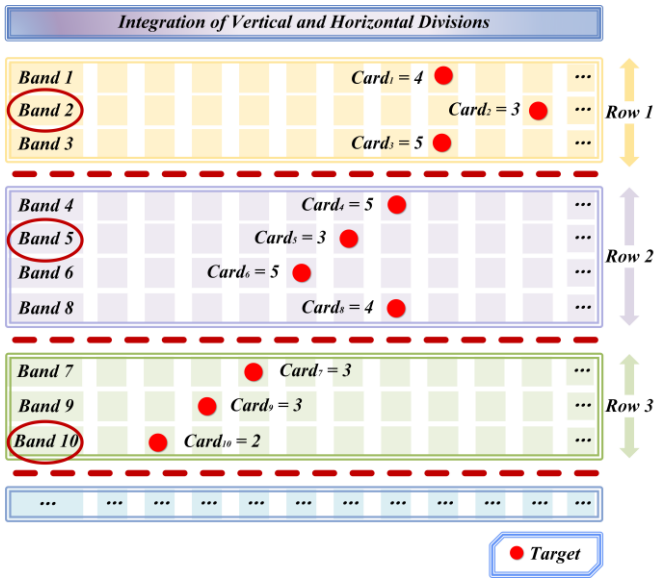


Fig. 5. Chessboard-shaped topological framework.

grid-shaped point set have small numerical differences and high similarity; hence, they are more likely to be considered as belonging to the same class of objects. The two classes of pixels, target and background, are inevitable to be divided into separate point sets due to their distinct differences in data features. Furthermore, the relative position and density of these two classes of pixels in topological space are significantly different, resulting in significant differences in the cardinalities of the point sets in which they are located. The observation of Fig. 3 reveals that, with the exception of very few cases, the grid-shaped point sets containing targets have fewer pixels in each band. In other words, the cardinalities of the grids in which the targets are located are relatively small. Those pixels that are similar to the target in terms of data features exhibit similar intuitive properties in the topological space, resulting in a relatively small quantity and density on the chessboard. In contrast, the cardinalities of the grids in which the background is located are larger, indicating a relatively large quantity and density. For the same band, the cardinalities corresponding to the background are larger than the target. As a result, the division of the chessboard intuitively presents the differences in quantity and density of these two classes of pixels. Therefore, the rearrangement of the pixels can be used to distinguish the classes to which they belong. With the spatial disassembly, the neat and ordered form is reflected in the fact that, from left to right, the center values of point sets are increasing at equal intervals and that pixels with a degree of similarity in data features are in the same point set. As a result, various classes of pixels with different data features, which were difficult to separate in the original space due to their disordered arrangement, become ordered and easy to separate in the topological space, providing a reliable and solid basis for subsequent retrieval of pixels belonging to the same class as the given query.

B. Development of Distance on Chessboard

With the generation of chessboard-shaped topological spaces, the difficulty of retrieval against objects of interest is drastically reduced. There is an urgent need to address the core requirement of highlighting the target while suppressing the background by performing the algorithmic output for HSIs. As mentioned earlier, the differences in relative position and density between the target and the background in the topological space are directly reflected in the topological relationships between the separate point sets. Based on such a fact, the output of the proposed method makes full use of the unique structural advantages of the chessboard-shaped framework in two ways. First, the interval spans of values for the vertical and horizontal divisions are equal, and hence, the density of each grid-shaped point set is a direct reflection of the outliers of the pixels within it. Second, the shape of the chessboard itself is quite regular and corresponds perfectly to the coordinate system, making it easy to track and locate each pixel within it. As a result, the latent and hidden data features of various classes of objects are transformed into intuitive properties and the differences between target and background are easily measured, which facilitates the design of detection output.

In an HSI, all classes are treated as background, except for objects of interest that belong to the same class as the given query, which are treated as target. Thus, the information retrieval task essentially focuses on only two classes: the target and the background. Given that pixels in an HSI exist in the form of spectra, the differences in data features between the target and the background are essentially numerical differences between the two classes of spectra in specific bands. The vertical division of the chessboard is implemented by spatial disassembly until traversing all bands of the HSI with a single-band image as the subject each time, enabling the target and background pixels to be assigned to different columns in specific bands, which ensures a separation of the two classes.

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Considering the inherent spectral redundancy of HSIs, it is inevitable that the target and the background do not exhibit strong separability in all bands. Accordingly, with a given target spectrum as reference information, the horizontal division of the chessboard is integrated with the vertical division to adaptively select the optimal separable bands between the target and the background.

For a given target spectrum $\mathbf{t}$, its distribution in the topological space is tracked. With cardinalities in all bands recorded in $\mathbf{Card}_t$, a set of cardinalities corresponding to each row is found according to the aforementioned band indexes recorded for the given target. In Fig. 5, for the first row of the chessboard, the cardinalities corresponding to $\mathbf{t}$ are Card_1, Card_2, and Card_3. For the second row, the cardinalities of $\mathbf{t}$ are Card_4, Card_5, Card_6, and Card_8. For the third row, the cardinalities of $\mathbf{t}$ are Card_7, Card_9, and Card_{10}. Given that the cardinality of each grid-shaped point set reflects the degree of outliers of the pixels within it, the smaller the cardinality, the more separable they are. It can be concluded that the given target exhibits stronger separability from the background on those bands with smaller cardinalities. Therefore, for each row of the chessboard, the band index corresponding to the minimum in a set of cardinalities is selected as the optimal separable band, which is marked with a red circle in Fig. 5. On this basis, a breakthrough in the design of the detection output is to explore the measurement of target-background separation based on the optimal separable bands.

As described above, as a given query, the information of the target spectrum is fully exploited to select the optimal separable bands, each of which corresponds to each row of the chessboard. The selection process reduces the inherent spectral redundancy, which streamlines the initial chessboard-shaped topological framework for information retrieval tasks. In the streamlined chessboard, the differences between target and background in HSIs are further amplified to achieve accurate separation, and a separation measurement called "distance on chessboard" is developed for the detection output. For a testing pixel and the given target, the calculation of the distance on chessboard is accomplished by tracking and recording the distribution of the two in the streamlined chessboard-shaped topological space.

Specifically, $\mathbf{x} \in \mathbb{R}^{L\times 1}$ denotes a testing pixel in the HSI, the column to which $\mathbf{x}$ is assigned is tracked according to the interval in which its value lies in the optimal separable band for each row. The locations of the corresponding grid-shaped point sets are recorded row by row in $\mathbf{Coor}_x = [Coor_{x1}, \ldots, Coor_{xhor}]^{\mathrm{T}} \in \mathbb{R}^{\mathrm{hor}\times 1}$ to obtain the coordinates on the chessboard. For the given target $\mathbf{t}$, the same operation is performed to obtain its coordinates $\mathbf{Coor}_t = [Coor_{t1}, \ldots, Coor_{thor}]^{\mathrm{T}} \in \mathbb{R}^{\mathrm{hor}\times 1}$ according to its previously recorded distribution on the chessboard. The coordinates of the testing pixel and the given target are subtracted, and then, the subtractive vector is normed with ℓ_1. Also, the distance on chessboard between the testing pixel and the given target is defined as follows:

$$\mathrm{DOC} = \frac{\|\mathbf{Coor}_x - \mathbf{Coor}_t\|_1}{\mathrm{hor}}. \tag{1}$$

The above calculation for the distance on chessboard is shown in Fig. 6. Ultimately, the negative exponential function is adopted for the detection output to comply with the core requirement of highlighting the target while suppressing the background

$$D_{\mathrm{CTTD}}(\mathbf{x}) = \exp(-\mathrm{DOC}). \tag{2}$$

With the detection result, we have the following criteria for judgment:

$$D_{\mathrm{CTTD}}(\mathbf{x}) \begin{cases} > \eta \rightarrow & \text{target} \\ < \eta \rightarrow & \text{background} \end{cases} \tag{3}$$

where η is a set threshold. For the testing pixel $\mathbf{x}$, if $D_{\mathrm{CTTD}}(\mathbf{x}) > \eta$, then $\mathbf{x}$ is more likely to be judged as belonging to the same class as the given target $\mathbf{t}$. Conversely, if $D_{\mathrm{CTTD}}(\mathbf{x}) < \eta$, then $\mathbf{x}$ is more likely to be judged as belonging to a different class from $\mathbf{t}$, i.e., the background.

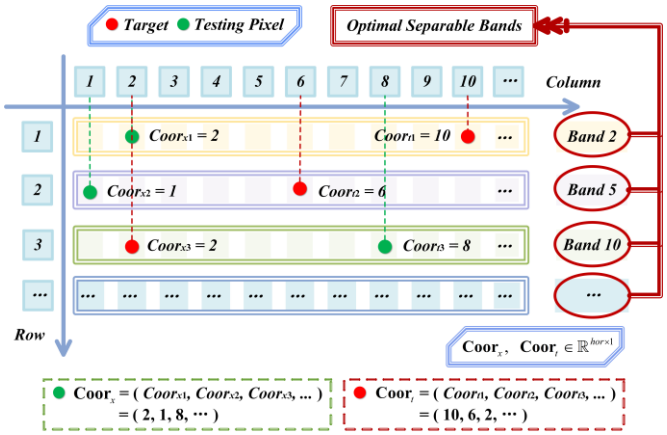


Fig. 6. Illustration for the calculation of distances on chessboard.

C. Information Retrieval in Topological Space

The application of the theories of point-set topology provides a viable solution to the information retrieval task. The process of the information retrieval task can be divided into two phases.

1) *Phase 1:* A specific form of framework is constructed to generate a topological space in which data points belonging to various classes of objects are efficiently separated.

2) *Phase 2:* An information retrieval task is performed in the topological space, using a given query as reference information to design the algorithmic output to meet specific retrieval requirements.

Given that hyperspectral target detection is refined into an information retrieval task in this article, the two stages described above correspond to two phases in the process of the proposed method, which meets exactly two phased goals of detection.

1) *Goal 1:* To separate the target and the background.

2) *Goal 2:* To highlight the target while suppressing the background.

In the first phase, for an HSI being processed, vertical and horizontal divisions of the chessboard are implemented to achieve both spatial and spectral disassembly. First, with the same interval span of values, the minimum–maximum interval of an image in a certain band is divided equally into multiple parts, each of which corresponds to a column. In this process, all pixels in the current band are assigned to the corresponding columns and grid-shaped point sets are generated accordingly. The above operation is performed for all bands of the HSI to complete the vertical division. In addition, the cardinalities of

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Algorithm 1 TD-ChessBoard ($\mathbf{X}$, $\mathbf{t}$, ver, hor)

Input: $\mathbf{X} = [\mathbf{X}_1; \ldots; \mathbf{X}_L] \in \mathbb{R}^{L\times N}$, where $\mathbf{X}_i \in \mathbb{R}^{1\times N}$-the HSI being processed, $\mathbf{t} = [t_1, \ldots, t_L]^{\mathrm{T}} \in \mathbb{R}^{L\times 1}$-the given target spectrum, ver-the number of intervals for vertical division, hor-the number of intervals for horizontal division.

Output: $\mathbf{Card}_t \in \mathbb{R}^{L\times 1}$-the cardinalities on the chessboard recorded for $\mathbf{t}$, $\mathbf{Loca}_t \in \mathbb{R}^{L\times 1}$-the locations on the chessboard recorded for $\mathbf{t}$, $\boldsymbol{TD-ChessBoard}$-the topological framework.

Initialization: $\mathbf{Card}_t = [Card_1, \ldots, Card_L]^{\mathrm{T}} \in \mathbb{R}^{L\times 1}$, $\mathbf{Loca}_t = [Loca_1, \ldots, Loca_L]^{\mathrm{T}} \in \mathbb{R}^{L\times 1}$, $\boldsymbol{ChessVer} = [\boldsymbol{ChessVer}_1; \ldots; \boldsymbol{ChessVer}_L] \in \mathbb{R}^{L\times ver}$, where $\boldsymbol{ChessVer}_i \in \mathbb{R}^{1\times ver}$, $\boldsymbol{ChessHor} = \emptyset$.

1: Perform the vertical division for spatial disassembly by starting a traversal;

for $i = 1 : L$ **do**
2: The minimum-maximum interval of $\mathbf{X}_i$ is divided equally into ver parts, which correspond to ver columns.
3: The central values of all columns in the current band are recorded in $\boldsymbol{ChessVer}_i$.
4: The column to which $\mathbf{t}$ is assigned is tracked according to the interval in which t_i lies in the current band. And the cardinality and location of the corresponding grid-shaped point set are recorded in $Card_i$ and $Loca_i$ respectively;
5: **end for (step 1)**

6: Perform the horizontal division for spectral disassembly. The minimum-maximum interval of $\mathbf{t}$ is divided equally into hor parts, which correspond to hor rows.
7: Each band of $\mathbf{t}$ is assigned to the corresponding row according to the interval in which its value lies, and each row records a set of band indexes.
8: The vertical and horizontal divisions are integrated to construct the topological framework by starting a traversal;

for $j = 1 : hor$ **do**
9: The band indexes of $\mathbf{t}$ in the current row are recorded in $\boldsymbol{ChessHor}_j$;
10: The band indexes of the given target in the current row $\boldsymbol{ChessHor}_j$ are added to $\boldsymbol{ChessHor}$;
11: With $\boldsymbol{ChessVer}$, the corresponding vertical divisions are found according to $\boldsymbol{ChessHor}_j$;
12: **end for (step 8)**

13: The vertical and horizontal divisions of the chessboard, $\boldsymbol{ChessVer}$ and $\boldsymbol{ChessHor}$, are integrated into $\boldsymbol{TD-ChessBoard}$;
14: **return** $\mathbf{Card}_t$, $\mathbf{Loca}_t$, $\boldsymbol{TD-ChessBoard}$

the grid-shaped point sets into which the given target spectrum is assigned are recorded for all bands. Second, with the same interval span of values, the minimum–maximum interval of a given target spectrum is divided equally into multiple parts, each of which corresponds to a row. In this process, all bands of the given target are assigned to the corresponding rows, and the band indexes in each row are recorded to complete the horizontal division. Third, the recorded band indexes are utilized as intermediate links to integrate the vertical and horizontal divisions. The corresponding vertical divisions are found through the band indexes recorded in each row, and a chessboard-shaped framework is constructed to generate the topological space. The detailed and specific steps of Phase 1 are described in Algorithm 1.

In the second phase, with the priori knowledge as a reference, the chessboard topology is exploited to select the optimal separable bands to calculate the distance on chessboard between the testing pixel and the given target spectrum for the detection output. First, the distribution of the given target in the topological space is tracked. According to the cardinalities of the given target in all bands recorded in the vertical division and the band indexes corresponding to rows recorded in the horizontal division, a set of cardinalities of the grid-shaped point sets into which the given target is assigned in each row is found. Second, the band index corresponding to the minimum of a set of cardinalities in each row is found, which is selected as the optimal separable band, and the chessboard-shaped topological framework is streamlined accordingly. Third, the coordinates of the testing pixel and the given target in the streamlined chessboard are recorded and subtracted, and the subtractive vector is utilized to calculate the distance on chessboard for the detection output. Algorithm 2 describes the implementation steps of Phase 2 in detail.

The proposed method includes two free parameters, ver and hor, which determine the vertical and horizontal divisions of the chessboard, respectively. Therefore, the setting of the parameters has a direct impact on the construction of the topological framework and thus on the execution of the subsequent information retrieval tasks. In Section III, various aspects of detection effect and computational efficiency are discussed to comprehensively evaluate the impact of the parameters on the proposed method.

III. Experimental Results and Analysis

In order to verify the validity of the proposed method, real HSIs with a complex distribution of imaging scenes are selected as experimental datasets. In addition, both qualitative and quantitative analyses are carried out to comprehensively evaluate experimental results. Section III-A provides an overview of the HSI datasets, evaluation indicators, software platform, and hardware configuration. Section III-B analyzes the impact of the free parameters on the proposed method in terms of detection effect and computational efficiency. Section III-C compares CTTD with other classic and state-of-the-art methods to demonstrate superiority.

A. Experimental Preparation

1) Datasets: Five publicly available benchmark HSIs are utilized for parametric discussions and performance comparisons, and the detailed information is listed in Table I.

1) The Airborne Visible/Infrared Imaging Spectrometer (AVIRIS)-1 dataset was acquired with the AVIRIS [53]. There are 100 × 100 pixels of an area in San Diego being collected. The spatial resolution is 3.5 m. Also, the spectrum spans a wavelength range of 370–2510 nm, from which 224 spectral channels are collected. After removing the absorbing and low signal-to-noise ratio (SNR) portions, 189 bands are retained. The objects of interest are three airplanes in three positions, occupying 20, 22, and 22 pixels.
2) The Hyperspectral Digital Imagery Collection Experiment (HYDICE) dataset was acquired through the HYDICE [54]. It contains 80 × 100 pixels acquired with a spatial resolution of approximately 2 m covering an area of MI, USA. There are 175 bands with good quality that are reserved for experiments. The ground

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Algorithm 2 CTTD (**X**, **target**, $\mathbf{Card}_t$, $\mathbf{Loca}_t$, $\boldsymbol{TD-ChessBoard}$)

Input: $\mathbf{X} \in \mathbb{R}^{L\times N}$-the HSI being processed, $\mathbf{t} \in \mathbb{R}^{L\times 1}$-the given target spectrum, $\mathbf{Card}_t \in \mathbb{R}^{L\times 1}$-the cardinalities on the chessboard recorded for **t**, $\mathbf{Loca}_t \in \mathbb{R}^{L\times 1}$-the locations on the chessboard recorded for **t**, $\boldsymbol{TD-ChessBoard}$-the constructed topological framework.

Output:d $= [d_1, \ldots, d_N] \in \mathbb{R}^{1\times N}$-detection results.

Initialization: $\mathbf{OSB} = [OSB_1, \ldots, OSB_{hor}]^{\mathrm{T}} \in \mathbb{R}^{hor\times 1}$, $\mathbf{Coor}_t = [Coor_{t1}, \ldots, Coor_{thor}]^{\mathrm{T}} \in \mathbb{R}^{hor\times 1}$.

1: Select the optimal separable bands by starting a traversal;

for $i = 1 : hor$ **do**
2: With $\boldsymbol{TD-ChessBoard}$, find the band indexes of the given target recorded in $\boldsymbol{ChessHor}_i$ for the current row;
3: With $\mathbf{Card}_t$ and $\boldsymbol{ChessHor}_i$, find the corresponding set of cardinalities for the current row and record them in $\mathbf{Card}_{t_i}$;
4; Find the minimum in $\mathbf{Card}_{t_i}$ and the corresponding band index OSB_i;
5: With $\mathbf{Loca}_t$ and OSB_i, obtain the coordinates of the given **t** in the current row,
$Coor_{ti} = \mathbf{Loca}_t(OSB_i)$;
6: **end for (step 1)**

7: Perform the information retrieval task by starting a traversal;

for $j = 1 : N$ **do**
8: $\mathbf{x}_j \in \mathbb{R}^{L\times 1} \leftarrow j$-th testing pixel of **X**;
9: Initialize the coordinates of the current testing pixel on the chessboard,
$\mathbf{Coor}_x = [Coor_{x1}, \ldots, Coor_{xhor}]^{\mathrm{T}} \in \mathbb{R}^{hor\times 1}$;
10: Obtain the coordinates of $\mathbf{x}_j$ on the chessboard by starting a traversal;

for $k = 1 : hor$ **do**
11: Find the optimal separable band corresponding to the current row, $OSB_k = \mathbf{OSB}(k)$;
12: The column to which $\mathbf{x}_j$ is assigned is tracked according to the interval in which $\mathbf{x}_j(OSB_k)$ lies in the current optimal separable band. And the location of the corresponding grid-shaped point set is recorded in $Coor_{xk}$;
13: **end for (step 10)**

14: A negative exponential function is adopted for the detection output,
$d_j = \exp(-DOC)$, where DOC is the distance on chessboard between $\mathbf{x}_j$ and **t**,
$DOC = \frac{\|\mathbf{Coor}_x - \mathbf{Coor}_t\|_1}{hor}$;
15: **end for (step 7)**

16: **return d**

truth is labeled with vehicles distributed in ten positions, each of which occupies 1–4 pixels.

3) The Los Angeles dataset was acquired by AVIRIS imaging of an area in Los Angeles [55]. The spatial dimension consists of 100 × 100 pixels with a resolution of 7.1 m. Also, the spectral dimension consists of 205 bands. The objects of interest are airplanes in two positions, occupying 43 and 44 pixels.
4) The Gulfport dataset is also named after the capture location [56]. It was acquired via AVIRIS and has a spatial size and a resolution of 100 × 100 pixels and 3.4 m, respectively. There are 191 bands in the spectral dimension reserved for experiments. The ground truth shows three airplanes to be detected with sizes of 10, 11, and 39 pixels.
5) The Cri dataset was acquired using the Nuance Cri hyperspectral sensor [57]. It images several rocks in a grassy area. The spatial size is 400 × 400 pixels. The spectral resolution is 10 nm and 46 bands are reserved. There are ten stones in different locations identified as objects of interest to be detected.

2) Evaluation Indicators: In addition to the colored detection maps that support direct visual observation, three widely used indicators are adopted to evaluate experimental results.

1) The receiver operating characteristic (ROC) curve illustrates the relationship between probabilities of detection (PDs) and FARs under a set of thresholds on the axis [58]. A prominent detection method exhibits high PDs with low FARs, and hence, the more the ROC is skewed toward the top-left corner, the better the effect. This curve also reflects the sensitivity of the detection method to the selection of the threshold.
2) The area under the curve (AUC) is calculated by integrating the ROC [59]. As it is difficult to distinguish subtle differences between algorithms by visual observation of ROC curves, the AUC provides a more accurate and reasonable quantifiable indicator for evaluation. For ROC curves in the same axis, the larger the AUC, the better the detection capability of the corresponding algorithm.
3) The separability map illustrates the statistical range of detection results and visualizes the target-background separation [60]. With locations of the target and the background in the ground truth, the corresponding two sets of values in the detection results are plotted in distinguishable colored boxes. The slighter the intersection between the two colored boxes, the more accurately the algorithm separates the target.

3) Equipment and Configuration: The hardware configuration is a Windows 11 computer with a 12th Gen Intel[1] Core[2] i7-1260 CPU @ 2.10 GHz and 16.0 GB RAM. Also, the software platform on which experiments are conducted is MATLAB R2014a.

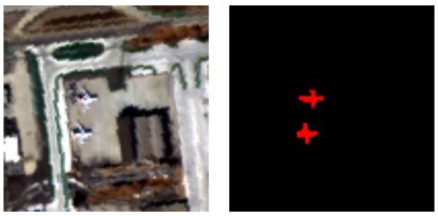
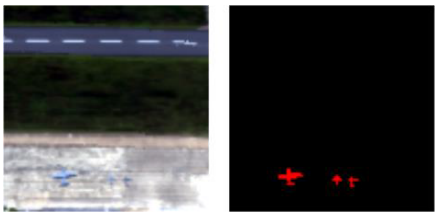
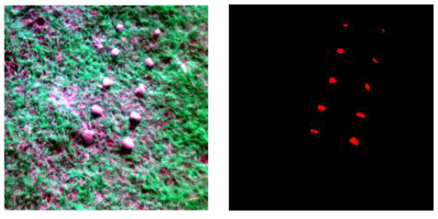
B. Discussion on Parameters

1) Detection Effect: By cascading the steps described in Algorithms 1 and 2, the complete algorithmic process of CTTD includes two free parameters for the input. For an HSI being processed, ver acts in the vertical division for spatial disassembly, while hor acts in the horizontal division for spectral disassembly. As mentioned earlier, the method proposed in this article finds the solutions for hyperspectral target detection based on data features of HSIs themselves rather than specific model assumptions As the information listed in Table I, the experimental datasets vary in spatial and spectral dimensions, so setting ranges are selected for both free parameters to develop the discussion. Also, AUC is adopted as a quantitative indicator to evaluate the detection

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TABLE I
DETAILED INFORMATION OF EXPERIMENTAL DATASETS

Data Name	Experimental Data sets				
	Spectro-meter	Capture Location	Spatial Resolution	Spatial Size	Number of bands reserved
AVIRIS-1	AVIRIS	San Diego	3.5 m	100×100	189
					
HYDICE	HYDICE	Michigan	2m	80×100	175
					
Los Angeles	AVIRIS	Los Angeles	7.1m	100×100	205
					
Gulfport	AVIRIS	Gulfport	3.4m	100×100	191
					
Cri	Nuance Cri	Wuhan University	\	400×400	46
					

AVIRIS: Airborne Visible/Infrared Imaging Spectrometer
HYDICE: Hyperspectral Digital Imagery Collection Experiment

effect under different combinations of parameters. Given that the topological space cannot be constructed when ver and hor are equal to one, both parameter ranges are set to [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], whereby there are a total of 19 × 19 parameter combinations. Theoretically, there is

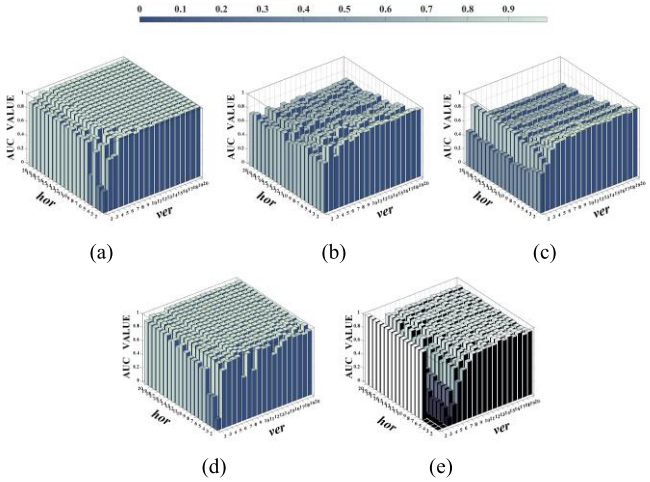


Fig. 7. Colored bar charts illustrating the relationship between combinations of ver and hor and the detection effect of CTTD on experimental datasets. (a) AVIRIS-1. (b) HYDICE. (c) Los Angeles. (d) Gulfport. (e) Cri.

no specific constraint between the settings of ver and hor and the number of pixels and bands in an HSI being processed. In the spatial disassembly, an image in a specific band of the HSI is extracted, and its minimum–maximum interval is divided equally into ver parts, which correspond to ver columns. This is not limited to the number of pixels in the HSI. Similarly, hor for spectral disassembly is not limited to the dimension of the given target spectrum, that is, the number of bands. For the impact of parameters on the proposed method, such a scheme of observing experimental results over a broad range of parameters is conducive to more reasonable and solid conclusions.

With bar charts colored according to the magnitude of values to plot AUCs obtained under different parameter combinations, Fig. 7(a)–(e) shows the detection effects on AVIRIS-1, HYDICE, Los Angeles, Gulfport, and Cri, respectively. Also, in order to explore recommendations for parameter settings for more diverse HSIs, Table II gives the optimal combination of parameters when AUC reaches the maximum on all experimental datasets. Conclusions can be drawn from the observation.

1) Overall, with the exception of a few combinations where both parameters are extremely small, the values of AUC in Fig. 7 increase gradually and fluctuate less as ver and hor increase. The gentle trend of the colored bar charts over a broad range indicates that the proposed CTTD is less sensitive to the varying parameters and exhibits robustness.
2) In Table II, it can be seen that on AVIRIS-1, HYDICE, and Los Angeles, AUC reaches the maximum with larger ver and smaller hor, indicating that the proposed CTTD excels with parameter settings that conform to such a pattern. In contrast, for Gulfport and Cri, the recommended setting for ver is significantly reduced, while for hor, it is increased. Therefore, for HSIs where the imaging scenes are relatively complex in distribution, as in the first three datasets, it is recommended that ver is increased and hor is decreased within a reasonable range. For HSIs with a more homogeneous distribution of imaging scenes like the fourth and fifth datasets, it is recommended to reduce ver and increase hor as appropriate.

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TABLE II
RECOMMENDED PARAMETER SETTINGS FOR CTTD ON EXPERIMENTAL DATASETS

Experimental Data Set	Evaluation Result (AUC)	Recommended Parameter Setting	
		ver	*hor*
AVIRIS-1	0.9984	13	2
HYDICE	0.9907	19	2
Los Angeles	0.9802	20	2
Gulfport	0.9935	6	9
Cri	0.9999	2	15

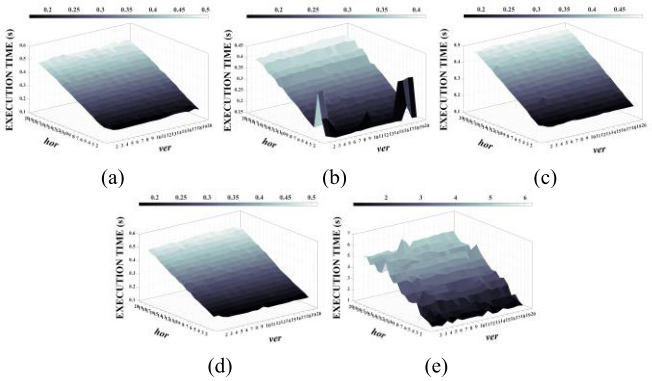


Fig. 8. Colored surfaces illustrating the relationship between combinations of ver and hor and the computational efficiency of CTTD on experimental datasets. (a) AVIRIS-1. (b) HYDICE. (c) Los Angeles. (d) Gulfport. (e) Cri.

The above parameter recommendations can be explained by the mechanism of the proposed CTTD. As mentioned in Section II-B, in an HSI, all classes are treated as background, except for objects of interest that belong to the same class as the given query, which are treated as target. Thus, the information retrieval task essentially focuses on only two classes: the target and the background. For HSIs with a relatively homogeneous distribution of imaging scenes, fewer classes of objects lead to a homogeneous distribution of the background. The spatial disassembly based on the chessboard should not be overly refined, and therefore, ver should be set smaller. For the spectral dimension, there is less interference from the homogeneous background distribution, resulting in more available bands for the target to be separated out. Hence, the spectral disassembly can be refined to accumulate the differences between these two classes of objects, and it is recommended that hor is set larger. As for HSIs with a more complex distribution of imaging scenes, the variety of objects leads to a complex background distribution as well. In this case, the chessboard-based spatial disassembly should be refined, so it is recommended that ver is set larger. In the spectral dimension, the interference caused by the complex background distribution results in fewer bands available to separate the target. An overly refined spectral disassembly can rather diminish the differences between the two classes, so it is recommended to set hor smaller.

2) Computational Efficiency: Fig. 8 shows the relationship between the execution time of CTTD and ver and hor to demonstrate the impact of the parameter settings on the computational efficiency of the proposed method. The execution time recorded covers Phases 1 and 2 of the proposed method. It is obvious that the colored surfaces representing time are generally sloping on all datasets. With the exception of local aberrations on HYDICE and Cri due to systematic errors caused by the equipment, the slopes are relatively smooth for other datasets. As hor increases, the time consumed by CTTD increases linearly and slowly. The computational cost of the autonomously developed distances on chessboard is quite low, whereby the linear complexity with which CTTD is equipped allows it to operate efficiently. Currently, hyperspectral datasets with large scales place higher demands on computational efficiency in practical applications, which poses a technical challenge for a large number of detection methods with expensive computational costs. Therefore, despite the need for a parameter tuning process, the proposed method is quite applicable to HSIs with variability to meet the requirements of high-precision detection in real scenes.

C. Comparison of Performance

1) Detection Effect: To further validate the prominent performance for the proposed method, CTTD is compared with other classic and advanced detection methods: MF [61], GLRT [26], CEM [62], ACE [24], signed ACE (S-ACE) [63], CSCR-based detector [39], SLRMD [40], and decomposition model with background dictionary learning (DM-BDL) [65]. On each dataset, the same given target spectrum is utilized as a priori knowledge for all participating algorithms to ensure a fair and equitable comparison. Considering that some of the aforementioned methods adopt the concentric dual-window detection mode, the window sizes are set according to the information provided in Section III-A about the objects of interest to achieve excellent detection results on all experimental datasets. The inner window for AVIRIS-1 is set to 5×5 and the outer window is 15×15. The outer and inner windows for HYDICE are set to 13×13 and 3×3, respectively. The outer and inner windows set for both Los Angeles and Gulfport are 17×17 and 7×7, respectively. The window sizes for Cri are 25×25 and 15×15. The same computer equipment and software platform are employed to conduct the experiments for comparison and the results are all evaluated under the same conditions.

By linearly stretching detection results of all the methods involved in the comparison to the range 0–255, Fig. 9 presents the detection maps colored according to the magnitude of values on experimental datasets. Careful observation reveals the following conclusions.

1) The detection maps of MF, CEM, CSCR, and SLRMD have higher values of the background component, and the contours of artificial buildings such as roads and houses are visually obvious. The high false alarms indicate poor suppression of the background.
2) The detection values of GLRT, ACE, and S-ACE are overall lower and more concentrated in the background component, with better suppression effects. However, the blurred contours of the objects of interest indicate a weak capability for highlighting the target.
3) In contrast, DM-BDL and CTTD not only have the stronger inhibitory effects on the background but also basically preserve the shapes and contours of the objects of interest. Moreover, the visual effects of the detection maps of CTTD are significantly superior and overall ahead of those of the other competitors.

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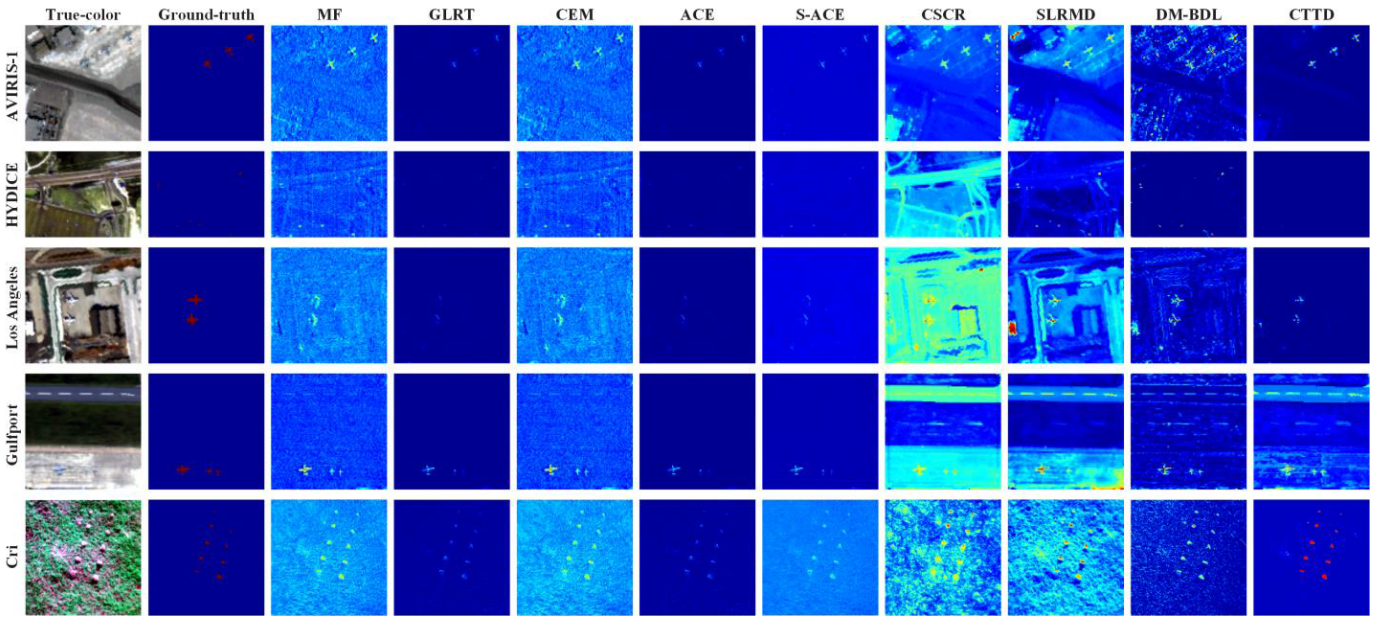


Fig. 9. Colored maps for comparison between detection methods on experimental datasets.

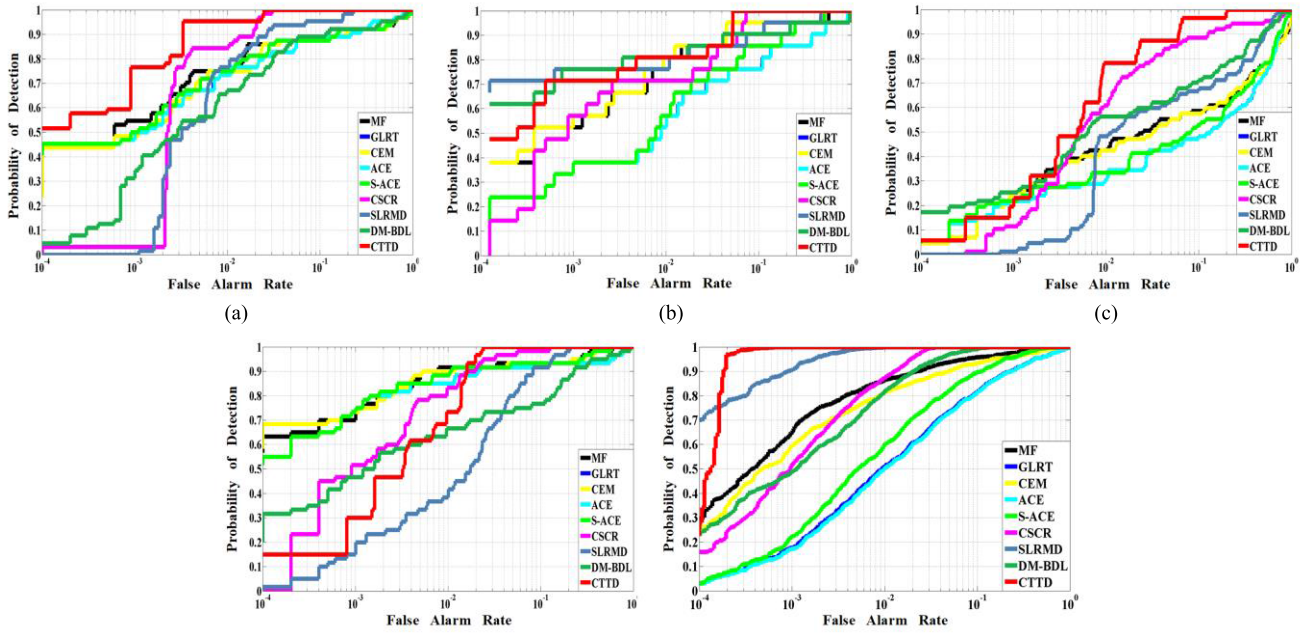


Fig. 10. ROC curves for comparison between detection methods on experimental datasets. (a) AVIRIS-1. (b) HYDICE. (c) Los Angeles. (d) Gulfport. (e) Cri.

The ROC curves for all methods are plotted in distinguishable colors on the same axis to observe the PDs at low FARs, and the ROC evaluations on AVIRIS-1, HYDICE, Los Angeles, Gulfport, and Cri are presented in Fig. 10(a)–(e), respectively. For quantitative analysis, AUC results of the detection methods on all datasets are summarized in Table III for comparison. On each dataset, the maximum of AUCs and the corresponding method are marked in bold. The following conclusions can be drawn.

1) On AVIRIS-1, Los Angeles, and Cri, the curves of CTTD are more to the upper left overall, exhibiting a clear advantage, while on HYDICE and Gulfport, despite a mediocre start, the PDs of CTTD both reach the highest after FAR > 0.1.
2) As a classic representation-based method, CSCR performs second only to CTTD, showing competitive detection effects. The ROC curves of the other methods involved in the comparison intersect with each other and show varying degrees of inferiority.
3) As shown in Table III, the AUC evaluation results of CTTD lead on all experimental datasets, demonstrating superior detection effects. The AUC results of CSCR are slightly inferior to those of CTTD, echoing the ROC evaluations in Fig. 10.

Fig. 11(a)–(e) presents the separability maps for all methods on AVIRIS-1, HYDICE, Los Angeles, Gulfport, and Cri. For each dataset, the left subfigure illustrates the statistical range

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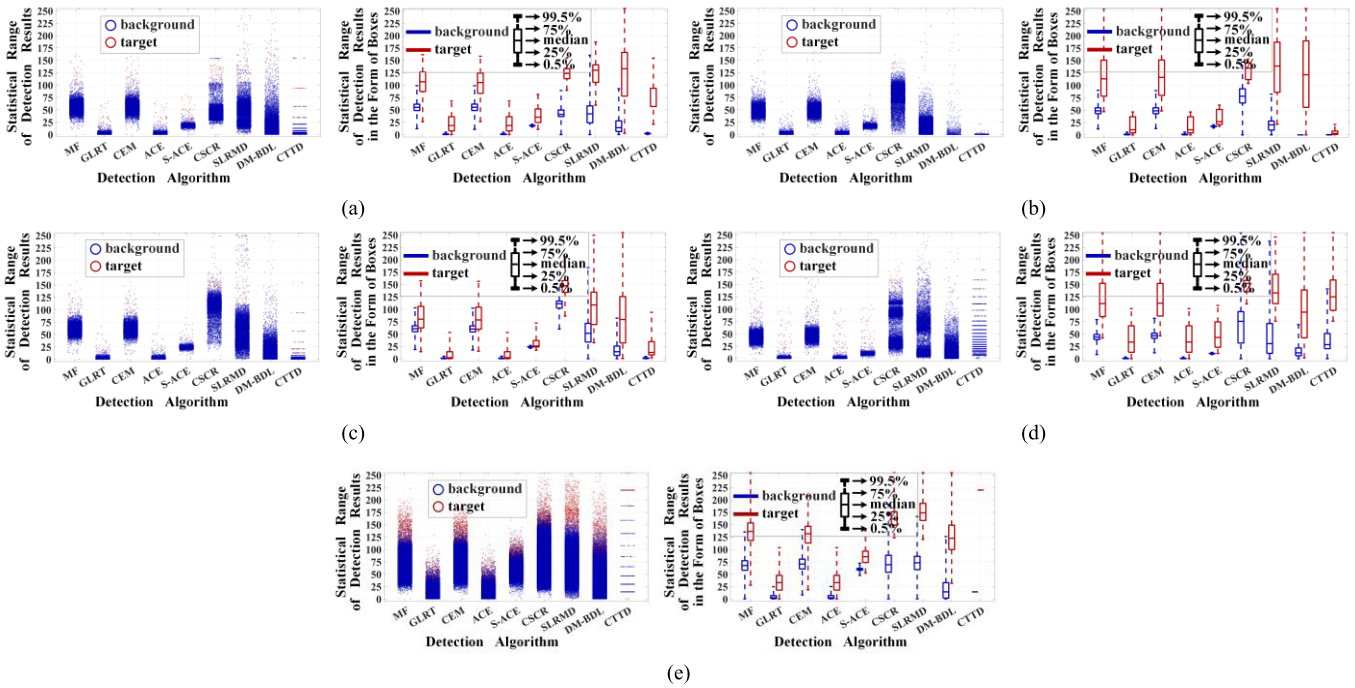


Fig. 11. Separability map for comparison between detection methods on experimental datasets. (a) AVIRIS-1. (b) HYDICE. (c) Los Angeles. (d) Gulfport. (e) Cri.

TABLE III

EVALUATION RESULTS FOR COMPARISON BETWEEN DETECTION METHODS ON EXPERIMENTAL DATASETS

Experimental Data Set	AVIRIS-1	HYDICE	Los Angeles	Gulfport	Cri
Method	**Evaluation Result --- AUC Value**				
MF	0.9340	0.9649	0.7580	0.9747	0.9820
GLRT	0.9379	0.8965	0.7268	0.9520	0.9246
CEM	0.9298	0.9692	0.7528	0.9750	0.9711
ACE	0.9379	0.8963	0.7265	0.9520	0.9234
S-ACE	0.9347	0.9476	0.7526	0.9741	0.9617
CSCR	0.9949	0.9858	0.9378	0.9929	0.9962
SLRMD	0.9838	0.9671	0.8547	0.9676	0.9997
DM-BDL	0.9374	0.9469	0.8720	0.9226	0.9925
CTTD	**0.9984**	**0.9907**	**0.9802**	**0.9935**	**0.9999**

of detection results, which is plotted in the form of boxes on the right side. The observations lead to conclusions as follows.

1) For GLRT, ACE, and CTTD, these three methods suppress the detection values of background locations to very low levels on AVIRIS-1, HYDICE, Los Ang, and Cri. On Gulfport, although the detection values of the background component for CTTD are slightly higher, CTTD is the most prominent of the three in highlighting the target.
2) In contrast, the detection values of the background component in MF, CEM, CSCR, SLRMD, and DM-BDL are relatively high and unconcentrated, which tends to raise false alarms. In addition, for S-ACE, the distribution of the statistical ranges of detection values is similar to that of ACE, except that the overall position is a little higher.
3) The intersection of the two boxes representing target and background in CTTD is quite slight, reflecting the strong capability of separating the two classes of objects. Furthermore, benefiting from the developed distance on chessboard, the detection values at background locations in CTTD are overall more concentrated and less likely to trigger false alarms, reflecting a low sensitivity to the selection of thresholds.
4) In combination with the ROC evaluation results in Fig. 10, it can be observed that when FAR < 0.01, the overall performance of CTTD is slightly inferior due to the intersection of statistical ranges of the values at target and background locations in the detection results. This corresponds to the slight intersection between the two sets of values in CTTD presented in Fig. 11.

Integrating the qualitative and quantitative evaluations, it can be observed that for HYDICE, CTTD can highlight the differences between the target and the background when the objects of interest occupy few pixels and are dispersed. For experimental datasets such as AVIRIS-1, Los Angeles, Gulfport, and Cri, when the objects of interest occupy a certain number of pixels and are contoured, CTTD basically preserves the spatial information of targets in the detection results while suppressing background interference to a very low level. These observations demonstrate that the proposed method is capable of amplifying and exploiting the differences in spatial and spectral dimensions between the target and the background for information retrieval tasks.

2) Computational Efficiency: For the detection methods involved in the comparison, they are all timed from the same start and the execution time covers the complete processing to ensure fairness. The execution time on all experimental datasets is recorded and summed in Table IV. It can be seen that on the same equipment and platform, except on Cri, CTTD is second only to traditional MF and CEM in terms of computational efficiency and far more efficient than the machine

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TABLE IV
EXECUTION TIME FOR COMPARISON BETWEEN DETECTION METHODS ON EXPERIMENTAL DATASETS

Method	MF	GLRT	CEM	ACE	S-ACE	CSCR	SLRMD	DM-BDL	CTTD
Data Set	**Execution Time of the Detection Algorithms --- Second**								
AVIRIS-1	0.1634	0.3325	0.0876	0.1940	0.3394	18.9759	51.1247	7.0170	0.1731
HYDICE	0.1198	0.2976	0.1045	0.1787	0.2011	12.8542	26.0163	6.3718	0.1630
Los Angeles	0.1595	0.3401	0.1575	0.3181	0.3721	19.2747	57.3543	7.7244	0.1795
Gulfport	0.1186	0.3930	0.0928	0.2073	0.3449	18.2737	60.6820	7.3120	0.2815
Cri	0.5889	1.1198	0.5054	0.9084	1.0287	575.7384	81.8002	82.7392	4.2598
Execution Time (sum)	1.1502	2.483	0.9478	1.8065	2.2862	645.1169	276.9775	111.1644	5.0569

learning-based methods of CSCR, SLRMD, and DM-BDL. The absence of any preparation of labeled samples or even a tedious training process makes the implementation of CTTD less demanding on memory occupation and time consumption. The proposed method with low computational cost is friendly for performing the data mining task of information retrieval in large-scale datasets such as HSIs.

IV. CONCLUSION

In this article, target detection in HSIs is refined as an information retrieval task to find solutions through point-set topology theories. To depart from traditional analytical modes that rely on specific model assumptions, we propose a novel detection method based on a chessboard-shaped topology. By autonomously designing a chessboard-shaped framework that is well-suited for high-dimensional data structures, HSIs are disassembled from both spatial and spectral dimensions to construct a topological space in which the differences between various classes of objects are sufficiently amplified to achieve accurate target-background separation. Performing the retrieval task in the topological space, we developed a distance measurement on chessboard for the algorithmic output to meet the retrieval requirements. The developed measurement is able to highlight the target while suppressing the background. The proposed CTTD is capable of adapting to real scenes with varying distribution and it has demonstrated superior detection effects. Furthermore, CTTD has linear time complexity, enabling efficient target localization on hyperspectral data with complexity. The competitive comprehensive performance of the proposed method promises to be advantageous in practical applications.

Despite the prominent overall performance, there is considerable scope for improvement in the proposed method. In the follow-up work, we will integrate the concentric dual-window detection model with the proposed method to maintain the spatial information of the original HSI data and experiment on hyperspectral data with diversity to make further improvements accordingly.

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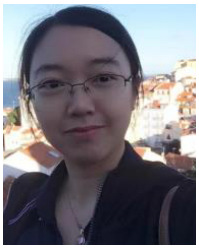
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